

CS102 PAIRED TECHNICAL INTERVIEW

This is the template for what you are expected to submit for the CS102 PAIRED Technical Interview.

Please make sure you have your items in this order when you submit so the we can grade it easier 🙏

Refer back to the Paired Technical Interview Print outs to the content you should submit.

In general, you should submit at least a number and some notes for the interviewer side, and your full code solution for the Candidate.

We encourage you to comment your code if you'd like!

Reginald Juance

Lilia Diusheveya

Interviewer Form

[Problem Given] Guess Number Higher or Lower

1) Problem Understanding: 4/4

- Lilia understands the problem to a binary search immediately off of the bat
- Ordered based on equal, greater than, or lower than
- Understands all the constraints given to her (size restrictions)

2) Communication & Collaboration: 2/4

- While she explained some things throughout the problem, like her thought process and reasoning behind Binary Search, she failed to engage with interviewer clearly
- At moments, she struggled to keep eye contact and social connections
- Too focused on coding, forgot to explain her entire thought process through the UMPIRE method

3) Implementation & Technical Depth: 4/4

- Time complexity is $O(\log n)$ --> natural to binary search
- Implementation is true and correct
- Updates range effectively based on guesses

4) Team Fit & Working Style: 3/4

- Lilia is a positive personality when going through the interview process
- She is incredibly receptive to hints, and is easy to work with
- She will undoubtedly work well in any conditions that she faces

Final Evaluation

- Score: 13/16
- Hiring Recommendation: Lean Hire

Final Decision

- While Lilia is completely capable in implementing the question successfully, she falls short in the team aspect a hiring manager is looking for.
She needs to learn to effectively communicate her thoughts when going through the interview process, or else this could raise concerns.
However,
throughout the interview, she was a positive person when acting under pressure and took into account any comments made. Her implementation was logically sound
and ensured a complexity of $O(\log n)$

Candidate Form

Code in Python:

```
# Two pointer/sliding window problem for left pointer & hash set to
prevent duplicates, ensures  $O(n)$  time complexity
# Sliding window to optimize time complexity  $O(n)$ 
def lengthOfLongestSubstring(s: str) -> int:
    left = 0 # left pointer to start window
    max_length = 0 # maximum substring length, initialize it as 0 &
update accordingly
    char_set = set() # ensures that string follows criteria
regarding valid strings, and keeps track of unique/non-duplicate
characters

    for right in range(len(s): # right pointer for pointer
        while s[right] in char_set: # remove invalid
characters or duplicates
            char_set.remove(s[left])
            left = left + 1 # move left pointer over to
slide

        char_set.add(s[right]) # add the current char to the
given set
        max_length = max(max_length, right - left + 1) # new
answer, accounting for current length

    return max_length
```